COURSE DESCRIPTION		
Course Code	24BST1101	Course Credit: 4
Course Name	Linear Algebra and Calculus	
Course Delivery Mode	TiC	
L-T-P-H-Z	3-1-0-0-0	
Pre-requisite	Nil	
Year of Introduction and Syllabus Version	2024 V1.0	

COURSE OBJECTIVES		Cognitive Level	Knowledge Category
1	To build a solid foundation in the fundamental concept of linear algebra.	L3	K3
2	To establish a solid foundation in multivariable functions and their behaviour through partial differentiation techniques.	L3	K3
3	To gain proficiency in utilizing multiple integrals to analyse and model spatially varying phenomena encountered in engineering disciplines.	L3	K3
4	To develop the ability to represent functions using series expansions such as Taylor series, Binomial series and Fourier series.	L3	K3

COURSE OUTCOMES (COs)	
After the successful completion of the course the students will be able to:	
CO1	Determine the rank of a matrix to investigate the existence and uniqueness of solutions, identify and analyse system behaviour under different conditions.
CO2	Apply fundamental principles of Eigen values, Eigen vectors, diagonalization and quadratic forms, enabling the analysis and interpretation of basic engineering problems.
CO3	Apply fundamental concepts such as partial derivatives, chain rule, local linear approximations and optimization techniques to analyse and solve problems involving functions of two variables.
CO4	Compute multiple integrals and apply them to find areas and volumes of geometrical shapes, mass and centre of gravity of plane laminas.
CO5	Describe functions using various series expansions such as Taylor series, Binomial series and Fourier series.

CO-PO MAPPING												
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	-	-	-	-	-	1	-	-	-
CO2	3	3	-	-	-	-	-	-	1	-	-	-
CO3	3	3	-	-	-	-	-	-	1	-	-	-
CO4	3	3	-	-	-	-	-	-	1	-	-	-
CO5	3	3	-	-	-	-	-	-	1	-	-	-
Correlation levels: 1- Low; 2-Medium; 3-High; No Correlation- "-"												

CO-PSO MAPPING			
	PSO1	PSO2	PSO3
CO1	2	-	-
CO2	2	-	-
CO3	2	-	-
CO4	2	-	-
CO5	2	-	-
Correlation levels: 1- Low; 2-Medium; 3-High; No Correlation- "-"			

COURSE CONTENTS

COURSE IMPLEMENTATION CLOCK (IN HOURS)	
Zeroth Module	**1 Hr.**
Course content delivery hours	**48 Hrs.**
Conclusive Module	**2 Hrs.**
Internal Test **(2 Assessment Tests of one and half hour duration)**	**3 Hrs.**

Zeroth Module: General Introduction to the Course		
● Course introduction with a thorough briefing of course objectives and course outcomes. ● An overview of the POs and PSOs that are mapped to the said course.	No ESE	1 Hr

Module-I		COs	Hrs.
Module Title	**System of Linear Equations**		
System of linear equations - Solution by Gauss elimination - Row Echelon form and Rank of a matrix - Fundamental theorem for linear systems of homogeneous and nonhomogeneous (without proof) - Homogeneous linear system - Non homogeneous linear system.		1	8
Module – II		**COs**	**Hrs.**
Module Title	**Matrix Eigen Value Problems**		
Eigen values and Eigen vectors - Properties of Eigen values - Diagonalization of matrices - Orthogonal transformation and orthogonal matrices - Quadratic forms and their Canonical forms.		2	8
Module – III		**COs**	**Hrs.**
Module Title	**Multivariable Calculus – Differentiation**		
Concept of limit and continuity of functions of two variables - Partial derivatives of first and higher order - Implicit partial differentiation - Local linear approximations - Chain rule for derivatives and partial derivatives - Relative maxima and minima of function of two variables (finding relative extrema only).		3	10

Module-IV		COs	Hrs.
Module Title	**Multivariable Calculus - Integration**		
Double integrals (Cartesian) - Reversing the order of integration - Change of coordinates (Cartesian to Polar) - Finding areas and volume using double integrals - Mass and centre of gravity of inhomogeneous laminas using double integral - Triple integrals - Volume calculated as triple integral - Triple integral in cylindrical and spherical coordinates (computations involving spheres and cylinders only).		4	12
Module-V		**COs**	**Hrs.**
Module Title	**Series Representation of Functions**		
Taylor series and Maclaurin series - Binomial series - Fourier series - Euler formulae - Convergence of Fourier series - Half Range sine and cosine series.		5	10

Conclusive Module: Contemporary topics/Industry-Academia interaction		
Guest lectures (online) from industry personnel / R&D organizations (National / International)	No ESE	2 Hrs.

TEXT BOOKS	
1	H. Anton, I. Biven and S. Davis, Calculus, 10th Edition, Wiley, 2015, ISBN: 978-81-265-5640-3.
2	E. Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley, 2015, ISBN: 978-81-265-5423-2.
3	B.S. Grewal, Higher Engineering Mathematics, 43rd Edition, Khanna Publishers, 2014, ISBN:.978-81-7409-195-5.
REFERENCE BOOKS / WEB REFERENCES	
1	M. D. Weir, J. Hass, F. R. Giordano, Thomas' Calculus, 11th Edition, Pearson, 2008, ISBN:978-81-317-1867-4.
2	G. B. Thomas, R. L. Finney, Calculus and Analytic Geometry, 9th Edition, Pearson, 2006, ISBN:81-7758-325-5.
3	H.K Dass, Advanced Engineering Mathematics, 1st Edition, S. Chand, Reprint 2010, ISBN: 81-219-0345-9.
4	B. Ram, Engineering Mathematics, 1st Edition, Pearson, 2012, ISBN: 81-317-9862-3.
5	F. Ayres, E. Mendelson, Schaum's Outlines Calculus, 4th Edition, Mc Graw Hill, 2000, ISBN: 0-07-116036-1.

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MODES OF EVALUATION		SCORE WEIGHTAGE / SPLIT MARKS
Continuous Internal Evaluation (CIE)		**40**
Internal Test - I		7.5
Internal Test - II		7.5
Task- works		15
Assignment	40%	
Class works	30%	
Problem solving	30%	
Attendance		10
End Semester Examination (ESE)		**60**
Total		**100**
End Semester Examination Pattern		
The question paper consists of two parts: Part A: 15 marks Part B: 45 marks Part A contains a total of 5 questions, 1 question from each module, having 3 marks. Part B contains a total of 10 questions, 2 questions from each module of which students should answer any one. Each question in Part B carries 9 marks and can have a maximum 2 sub-divisions.		

APPROVAL SUMMARY	
	Number and Date
Recommended by BoS	SCE/BoS/MA/009/23-24, 21/6/2024 SCE/BoS/CS/B.Tech/009/23-24, 20/6/2024
Approved by Academic Council	SCE/AC/013/2023-24 Dtd. 09/07/ 2024

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APPROVED BY ACADEMIC COUNCIL

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